Federated learning for quantifying racial disparities in kidney graft failure rates using US registry data from 29,468 patients across 149 transplant centers

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Background

Kidney transplant is a renal replacement therapy for eligible patients with end-stage renal disease (ESRD)[1–3]. Unfortunately, the racial disparities in receipt of a transplanted kidney are observed for the Black across states[4]. Black patients are recognized to have lower graft survival rates compared with White patients[5–8]. Therefore, there is a great need in understanding, quantifying, and reducing the racial disparities in access to transplant and post-transplant outcomes for kidney transplant patients.

Site of care has been considered as a major contributor to disparities in kidney transplants due to differences in time on the transplant waiting list, access to live donor kidney transplants, care coordination with the donor organ procurement system, and acute rejection rates[9]. Previous studies have suggested that access to kidney transplants across the US is associated with a person's geographic location[10], one of the social determinants of health (SDoH). The theory is that the SDoH, especially the racial residential segregation in the US exposes Black and White patients to different transplant centers with varying transplant care and uneven transplant outcomes[11,12].

To study the potential association between the site of care and racial disparity in kidney transplant graft failure with multi-site data, a counterfactual modeling method can be implemented[13]. By considering the underlying distribution in which patients of different races receive care, we can use a counterfactual model to simulate patients from a certain transplant center getting admitted to another. For example, with the proportions of White patients at the studied centers, we can simulate the reassignments for Black patients with the same proportions

within the hospitals hypothetically. Under this scenario, the Black patients are pretended to attend the hospitals in the same distribution as the White patients. The observed and counterfactual kidney graft failure rates can be calculated, and the hypothesis is that the counterfactual graft failure rates would be lower than the observed graft failure rates for the Black patients if there exist site-of-care-associated racial disparities for the Black patients in kidney transplants.

This counterfactual modeling has been recently implemented in the study by Asch et al (2020)[14] to explore racial disparities for patients hospitalized with COVID-19 infection. In this study, a Health Quality Forum certified generalized linear mixed model (GLMM) was fitted on a *centralized* multi-site data to characterize the odds of the event with the adjustments for both fixed (e.g., the effects of patient- and hospital-level factors) and random effects (e.g., hospital-specific effects). In studying the effect of site of care on an event rate (e.g., COVID-19 mortality rate, kidney graft failure rate), the hospital-specific effects are essential in the calculation of counterfactual event rate. However, the multi-site data sometimes cannot be centralized due to obviate privacy and confidentiality concerns. For example, in an international collaborative study within OHDSI network, it is challenging to share patient-level data across countries. Under this situation, the utilization of *centralized* modeling is not achievable, leading to challenges in estimating the hospital-specific effects. When the patient-level data are stored in a *decentralized* format (e.g., hospitals/institutions from multiple countries) and only aggregated data are allowed to be shared across hospitals, a *decentralized* algorithm for GLMM is in critical need.

We proposed a framework of federated learning algorithm to investigate the association between the site of care and racial disparities in kidney graft failure for Black patients. We termed this framework as dGEM-disparity, which stands for decentralized algorithm for Generalized linear mixed Effect Model for disparity quantification. dGEM-disparity is a federated learning framework to effectively integrate heterogeneous multi-site data by fitting generalized mixed effect models. It allows us to investigate the site-of-care-associated racial disparity by considering the patients who are hypothetically admitted to other hospitals. We believe that the proposed framework is a timely and significant contribution to the evidence-based quantification of racial disparity that is attributable to differential access to healthcare. The proposed framework facilitates multi-site collaboration and identifies hospital-level racial disparities, which will ultimately advance equity in human health.

Methods

There are three main steps in dGEM-disparity as shown in **Figure 1**. Let $\boldsymbol{K}$ denote the number of total hospitals. In Step I, each of the $\boldsymbol{K}$ sites estimates the common effect $\widehat{\boldsymbol{\beta}}_{\boldsymbol{k}}$ and variance $\boldsymbol{I}_{\boldsymbol{k}}^{-\mathbf{1}}\left(\hat{\boldsymbol{\beta}}_{\boldsymbol{k}}\right)$ with its own patient-level data. The common effects are combined to obtain a meta-estimate $\widehat{\boldsymbol{\beta}}$ of the common effect. The meta-estimate $\widehat{\boldsymbol{\beta}}$, then, is broadcasted to each site for the estimation of hospital effect $\hat{\gamma}_{\boldsymbol{k}}^{*}$ and its variance $\hat{\boldsymbol{s}}_{\boldsymbol{k}}^{*}$**.** In Step II, a hospital-level calibration through meta-regression is conducted with hospital-level characteristics $\boldsymbol{Z}_{\boldsymbol{k}}$ within each hospital and the estimated hospital effects $\{\hat{\gamma}_{\mathbf{1}}^{*}, \ldots, \hat{\gamma}_{\boldsymbol{K}}^{*}\}$ are shared among all hospitals. In Step III, each site

calculates the counterfactual event rates and the calibrated hospital effects $\{\hat{\gamma}_1^{**}, \ldots, \hat{\gamma}_K^{**}\}$ for disparity quantification.

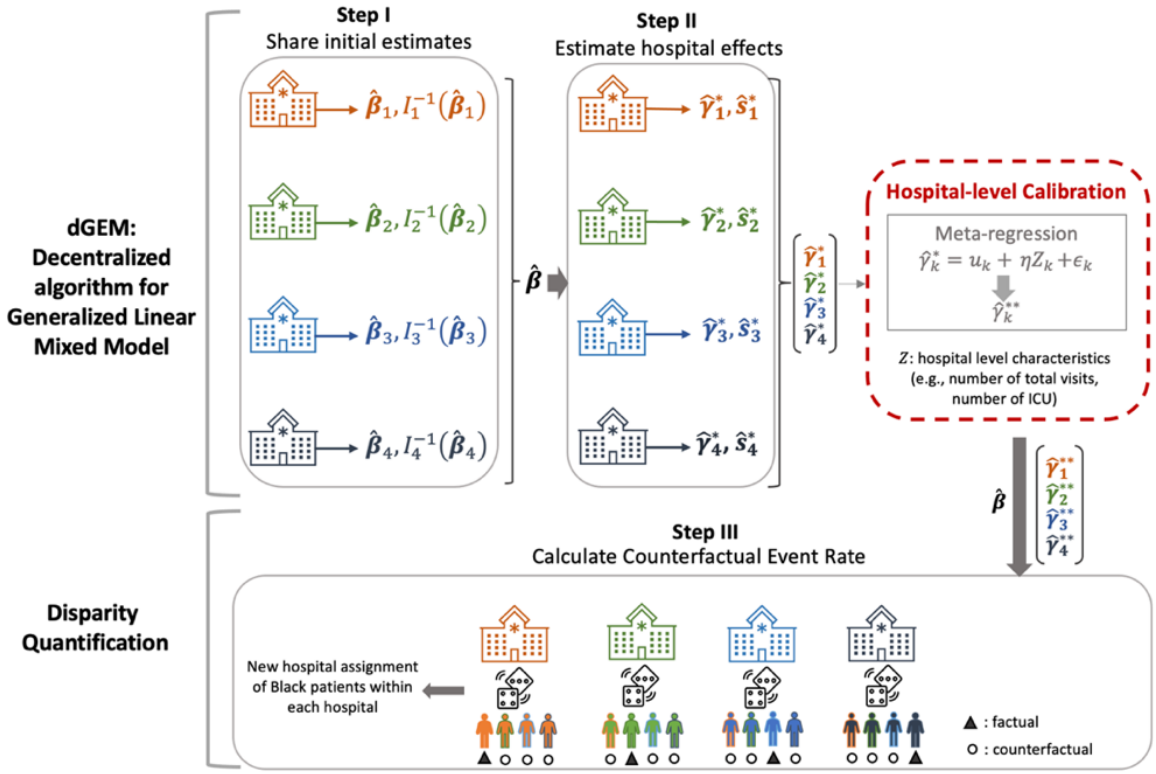


Figure 1. Workflow of the proposed decentralized algorithm for generalized mixed effect models framework for disparity quantification -- dGEM-disparity. This framework including three main steps: initial estimation of the common effect (i.e., association between outcome and covariates), estimation of hospital-effects with the meta-estimator combined from Step I, following with a hospital-level calibration, and the calculation of the counterfactual rates for disparity quantification.

The event rate in Step III can be calculated as

$$\hat{p}_k(\boldsymbol{X}_{kj}) = \text{expit}(\hat{\gamma}_k^{**} + \boldsymbol{X}_{k^*j}^{\boldsymbol{T}}\hat{\boldsymbol{\beta}}),$$

which is a factual event rate if $k = k^*$, and a counterfactual event rate if $k \neq k^*$ (i.e., pretending the patient from hospital k^* are admitted to hospital k). In the distributed format where patient-level data (i.e., $\boldsymbol{X}_{kj}$) cannot be shared across site, the calculation of the counterfactual event rate is achieved by transferring hospital-specific effects $\hat{\gamma}_k^{**}$ to protect the patient-level information.

The major aim of developing dGEM-disparity is to investigate what if the Black patients had been admitted to different hospitals for kidney transplant as the White patients. Therefore, we conducted a simulation with the centralized data to facilitate the above idea. In the simulation, we supposed that the Black patients had been admitted to hospitals in the same distribution as White patients while retaining their sociodemographic and clinical characteristics. The simulation procedure is depicted in a high-level conception in **Figure 2**. The proportions of White patients for all hospitals are calculated as $\{\widehat{w}_1, \ldots, \widehat{w}_K\}$. Within each hospital, the Black patients are randomly assigned to a hospital by a multinomial distribution with probabilities $\{\widehat{w}_1, \ldots, \widehat{w}_K\}$.

Hospital assignments are then used to calculate mortality risk estimates for each Black patient in their (counter)factual hospital, as shown in **Figure 2**. Within each hospital, the patient-level rate estimates of the Black patients with newly assigned hospital are summed up, and the overall counterfactual event rate estimate is obtained using these sums. The simulation procedure is replicated multiple times to quantify the uncertainty.

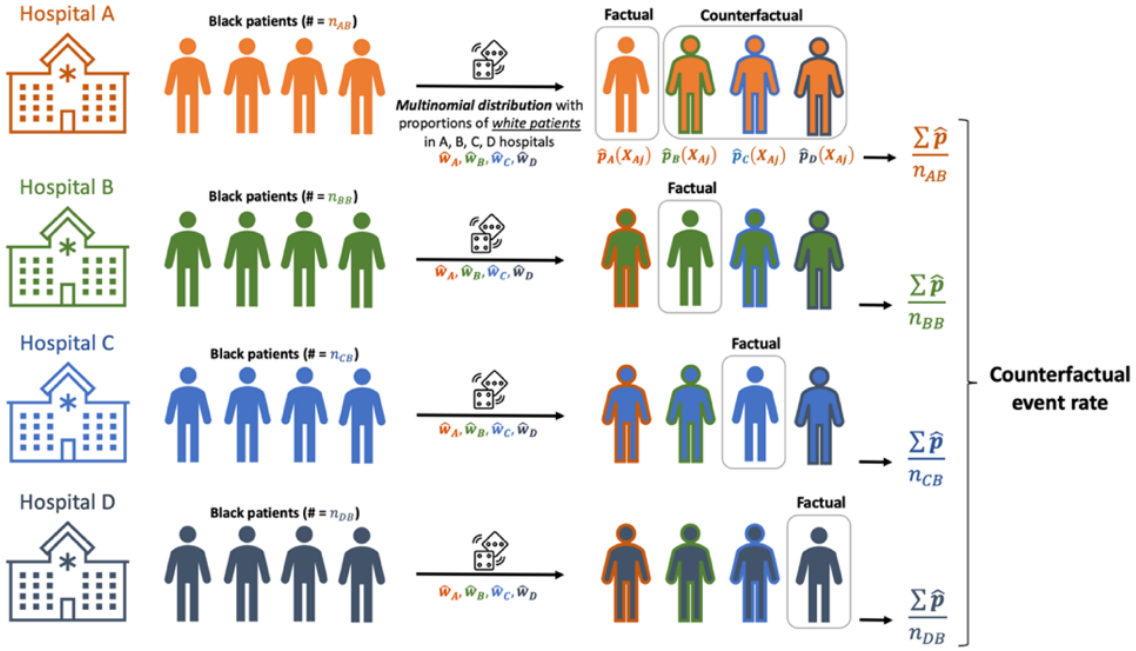


Figure 2. Overview of the calculation of counterfactual kidney transplant failure risks. Within each hospital, the black patients are reassigned with a hospital based on a multinomial distribution with the proportions of White patients in the hospitals. (Counter)factual rates are calculated within each site for Black patients, and then summed together to get the overall counterfactual event rate.

Results

The following Figure 3 summarizes the results using the registry data collected from the U.S. Organ Procurement and Transplantation Network (OPTN). OPTN is a unified transplant network established by Congress under the National Organ Transplant Act of 1982, which requires the submission of data on all solid-organ transplants in the United States. The OPTN national data include information on all transplant centers in the US. This dataset contains 29,468 Black and White adult recipients from 149 transplant centers who experienced a deceased donor kidney transplant between January 1, 2008, and December 31, 2011. 2,416 (8.2%) of the patients had either died or had graft failure (i.e., outcome) within one year after receiving a kidney transplant.

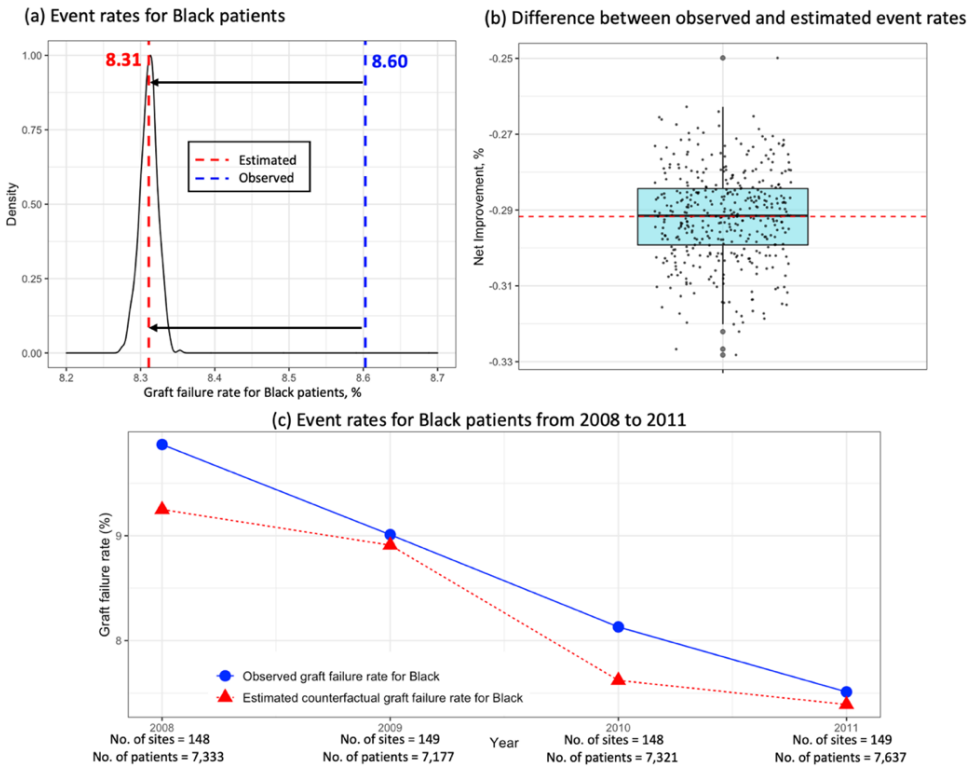


Figure 3. (a) Comparison between the estimated counterfactual event rate for Black (8.31% in blue) and the observed event rate (8.60% in red) for 29,468 patients. (b) Difference between the observed and estimated counterfactual rates from 400 replications, with an average decrease in rate by 0.29% (95% CI [0.27%, 0.32%]) (c) Results of the racial disparity in kidney transplant graft failure rate study with US registry data. The observed rate for Black patients and the estimated counterfactual event rate for Black patients with the proposed dGEM-disparity framework are compared within each year (from 2008 to 2011). At the bottom of each year, the total numbers of sites and patients for analysis are listed.

The observed graft failure rate for the Black patients was 8.60%. The estimated counterfactual graft failure rate decreased by 0.29% (95% CI, [0.27%, 0.32%]), down to 8.31%. This means that if Black patients had been admitted to the centers following the distribution of White patients, 29 fewer per 10,000 Black patients (95% CI: [27, 32]) would have died or had graft failure within one year after receiving a kidney transplant on average. By examining the difference over the time (Fig 3c), the graft failure rates fell in each year among the Black patients if the Black patients received their care in centers in the same distribution as White patients.

Conclusion

We presented the real-world application of the proposed dGEM-disparity framework. Our framework provides the estimates of fixed and random effects to further obtain the counterfactual event rates to investigate the association between the attending hospital and racial disparities. The results of the data analysis show if the Black patients in the study cohort are admitted to hospitals with the same distributions as the White patients, the event risk would

be reduced.

dGEM-disparity can be *generalized* to investigate other mediation effects associated with access to healthcare. For example, in the global combat against COVID-19, the disparity in health care of patients is considered as one of the most significant issues. Minority is broadly defined, not limited to race/ethnicity, and includes other disadvantaged subpopulations depending on socioeconomic status and age, among many other factors. Our proposed framework using dGEM-disparity is generally applicable to quantifying disparity-related differential access to healthcare for minority groups, and studying the factors that might relate to these disparities.

Currently, we are applying the dGEM part to study the COVID-19 hospital profiling across OHDSI network (https://github.com/ohdsi-studies/dGEMcovid, https://forums.ohdsi.org/t/call-for-participation-in-ohdsi-network-study-dgem-covid/16485). The federated algorithm has been successfully implemented across 15 sites from 4 countries using the R package (https://github.com/Penncil/pda) and a web-based software PDA-OTA (**p**rivacy-preserving **d**istributed **a**lgorithm **o**ver **t**he **a**ir, https://pda-ota.pdamethods.org/login) for secured sharing the aggregated data. We are planning to apply the dGEM-disparity by recruiting more sites who are also interested in the investigation on racial disparities across OHDSI networks.

Other than the racial disparities, other social determinants of health (SDoH) (e.g., income, education, insurance status, health literacy, and engagement) that contribute to the health disparities provide us opportunities to develop the federated learning algorithm to consider any types of fairness. Within the OHDSI networks, our future work will also investigate the outcome other than mortality, with a focus on utilizing real-world evidence to assess and develop potential intervention approached engaged in the organ transplant process.

In summary, we believe that the proposed framework is a timely and significant contribution to the evidence-based quantification of racial disparity that is attributable to differential access to healthcare. The proposed framework facilitates multi-site collaboration and identifies hospital-level racial disparities, which will ultimately advance equity in human health.

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